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1. AE 2026

A. Formulae

1.1.1

$$\Pr(M) + \Pr(E) - \Pr(ME) + \Pr(M + E)' = 1 \quad (1.1.1.1)$$

1.1.2

$$A + A' = 1, \quad AA' = 0 \quad (1.1.2.1)$$

$$\implies \Pr(A + A') = \Pr(1) = 1 \quad \Pr(AA') = \Pr(0) = 0 \quad (1.1.2.2)$$

1.1.3

$$\Pr(A + A') = \Pr(A) + \Pr(A') \quad (1.1.3.1)$$

1.1.4 Difference rule of probability

$$\Pr(AB') = \Pr(A) - \Pr(AB) \quad (1.1.4.1)$$

Solution: From (1.1.2.1)

$$A = A \cdot 1 = A(B + B') = AB + AB' \quad (1.1.4.2)$$

Also,

$$(AB)(AB') = 0 \quad (1.1.4.3)$$

From (1.1.3.1),

$$\Pr(A) = \Pr(AB + AB') = \Pr(AB) + \Pr(AB') \quad (1.1.4.4)$$

Rearranging, we get:

$$\Pr(AB') = \Pr(A) - \Pr(AB) \quad (1.1.4.5)$$

1.1.5 The Laplace transform is defined as

$$g(t) \xleftrightarrow{\mathcal{L}} G(s) = \int_0^\infty e^{-st} g(t) dt \quad (1.1.5.1)$$

1.1.6

$$g'(t) \xleftrightarrow{\mathcal{L}} sG(s) - g(0) \quad (1.1.6.1)$$

Solution: Applying integration by parts,

$$u = e^{-st}, dv = g'(t) dt \quad (1.1.6.2)$$

$$\implies \int_0^\infty e^{-st} g'(t) dt = [e^{-st} g(t)]_0^\infty - \int_0^\infty (-se^{-st}) g(t) dt \quad (1.1.6.3)$$

$$= -g(0) + s \int_0^\infty e^{-st} g(t) dt \quad (1.1.6.4)$$

yielding (1.1.6.1).

1.1.7

$$g''(t) \xleftrightarrow{\mathcal{L}} s^2 G(s) - sg(0) - g'(0) \quad (1.1.7.1)$$

1.1.8

$$e^{at}g(t) \xleftrightarrow{\mathcal{L}} G(s-a) \quad (1.1.8.1)$$

Solution:

$$\Rightarrow e^{at}g(t) \xleftrightarrow{\mathcal{L}} \int_0^{\infty} e^{-st} (e^{at}g(t)) dt \quad (1.1.8.2)$$

$$= \int_0^{\infty} e^{-(s-a)t} g(t) dt \quad (1.1.8.3)$$

yielding (1.1.8.1).

1.1.9 The unit step function is defined as

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} \quad (1.1.9.1)$$

1.1.10

$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s} \quad (1.1.10.1)$$

Solution:

$$\int_0^{\infty} e^{-st} dt = \left(-\frac{e^{-st}}{s} \right)_0^{\infty} = \frac{1}{s} \quad \Re(s) > 0 \quad (1.1.10.2)$$

1.1.11

$$tu(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s^2} \quad (1.1.11.1)$$

Solution: Differentiating (1.1.10.2) with respect to s results in (1.1.11.1).

$$\begin{aligned} \cos(at)u(t) &\xleftrightarrow{\mathcal{L}} \frac{s}{s^2 + a^2} \\ \sin(at)u(t) &\xleftrightarrow{\mathcal{L}} \frac{a}{s^2 + a^2} \end{aligned} \quad \Re(s) > a \quad (1.1.11.2)$$

Solution: From (1.1.8.1) and (1.1.10.1),

$$e^{jat}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s - ja}, \quad \Re(s) > a \quad (1.1.11.3)$$

$$\Rightarrow \cos(at)u(t) + j \sin(at)u(t) \xleftrightarrow{\mathcal{L}} \frac{s + ja}{(s - ja)(s + ja)} \quad (1.1.11.4)$$

$$= \frac{s}{s^2 + a^2} + j \frac{a}{s^2 + a^2} \quad (1.1.11.5)$$

resulting in (1.1.11.2).

1.1.12 Solve

$$\frac{d^2y}{dx^2} = -\alpha^2 y, \alpha \neq 0 \quad (1.1.12.1)$$

Solution: Applying the Laplace transform on both sides, and using (1.1.7.1),

$$s^2 Y(s) - sy(0) - y'(0) = -\alpha^2 Y(s) \quad (1.1.12.2)$$

Substituting

$$y(0) = \beta, y'(0) = \gamma \quad (1.1.12.3)$$

in (1.1.12.2),

$$Y(s) = \frac{s\beta + \gamma}{s^2 + \alpha^2} \quad (1.1.12.4)$$

$$= \frac{s\beta}{s^2 + \alpha^2} + \frac{\gamma}{s^2 + \alpha^2} \quad (1.1.12.5)$$

yielding

$$y(t) = \beta \cos(\alpha t) + \frac{\gamma}{\alpha} \sin(\alpha t) \quad (1.1.12.6)$$

from (1.1.11.2).

1.1.13 Euler's Method of Forward Differences. Numerically solve

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0 \quad (1.1.13.1)$$

Solution: By definition,

$$\left. \frac{dy}{dx} \right|_{x=x_0} = \lim_{h \rightarrow 0} \frac{y(x_0 + h) - y(x_0)}{h} \quad (1.1.13.2)$$

which can be expressed using the notation

$$y'_0 = \frac{y_1 - y_0}{h} \quad (1.1.13.3)$$

Similarly,

$$y''_n = \frac{y'_{n+1} - y'_n}{h} \quad (1.1.13.4)$$

From (1.1.13.3),

$$y'_{n+1} = \frac{y_{n+2} - y_{n+1}}{h}, y'_n = \frac{y_{n+1} - y_n}{h} \quad (1.1.13.5)$$

yielding

$$y''_n = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \quad (1.1.13.6)$$

upon simplification.

1.1.14 Kronecker delta function

$$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases} \quad (1.1.14.1)$$

1.1.15

$$f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{j2\pi n f_0 t} dt = \delta_n \quad (1.1.15.1)$$

1.1.16 The Fourier Series. Let

$$x(t) = \sum_{k=-\infty}^{\infty} x_0(t)(t - nT_0) \quad (1.1.16.1)$$

be a periodic signal with

$$x(t) = \begin{cases} x_0(t), & t \in \left(-\frac{T_0}{2}, \frac{T_0}{2}\right) \\ 0 & \text{otherwise} \end{cases}. \quad (1.1.16.2)$$

Then,

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_0 t}, \quad f_0 = \frac{1}{T_0} \quad (1.1.16.3)$$

Show that

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 t} dt \quad (1.1.16.4)$$

Solution: From (1.1.16.3),

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi n f_0 t} dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} \left(\sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t} \right) e^{-j2\pi n f_0 t} dt \quad (1.1.16.5)$$

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 t} dt = \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{j2\pi(k-n)f_0 t} dt \quad (1.1.16.6)$$

$$= \frac{1}{f_0} \sum_{k=-\infty}^{\infty} c_k \delta_{k-n} \quad (1.1.16.7)$$

upon substituting from (1.1.16.2) and (1.1.15.1), yielding (1.1.16.4).

1.1.17 The rectangular function. The unit rectangular function is defined as

$$\text{rect}(t) = \begin{cases} 1 & t \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ 0 & \text{otherwise} \end{cases} \quad (1.1.17.1)$$

1.1.18 Square Wave. Consider the Fourier series with

$$x_0(t) = \text{rect}(t), \quad T_0 = 2. \quad (1.1.18.1)$$

Show that

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} \quad (1.1.18.2)$$

Solution: Substituting $f_0 = \frac{1}{2}$ in (1.1.16.4),

$$c_n = \frac{1}{2} \int_{-1}^1 e^{-j\pi k t} dt \quad (1.1.18.3)$$

$$= \left[\frac{\sin(\pi k t)}{2\pi k} \right]_{-\frac{1}{2}}^{\frac{1}{2}} \quad (1.1.18.4)$$

The following code verifies the result in Fig. 1.1.18.1. Note that Fig. 1.1.18.1 is continuous while (1.1.18.1) is discontinuous. So (1.1.18.2) provides only an approximation. This is known as the Gibbs' phenomenon.

codes/Fourier.py

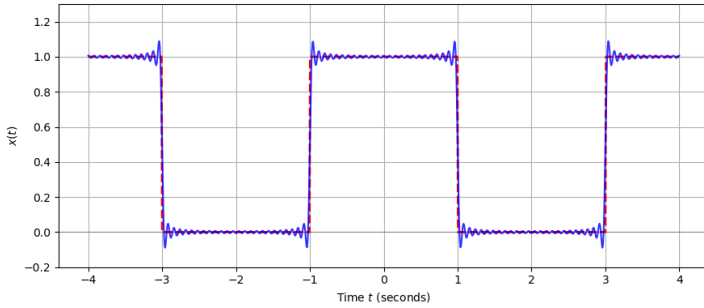


Fig. 1.1.18.1. Gibbs phenomenon

1.1.19 If

$$x(t) \xleftrightarrow{\mathcal{F}} c_n, \quad (1.1.19.1)$$

show that

$$x(t - t_0) \xleftrightarrow{\mathcal{F}} c_n e^{-j2\pi n t_0}, \quad (1.1.19.2)$$

Solution:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t - t_0) e^{-j2\pi n f_0 t} dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 (t+t_0)} dt \quad (1.1.19.3)$$

$$= e^{j2\pi n f_0 t_0} \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 t} dt \quad (1.1.19.4)$$

after a change of variables. Note that the limits of integration are unchanged due to the periodicity of $x(t)$. Transferring the exponential term in (1.1.19.4) to the L.H.S yields (1.1.19.2).

1.1.20 The standard 2-D basis vectors are defined as

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (1.1.20.1)$$

Also,

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.1.20.2)$$

1.1.21 For

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top, \text{rank}(\mathbf{A}) = 1. \quad (1.1.21.1)$$

where both u and v are n -dimensional real-valued non-zero column vectors.

Solution: Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}. \quad (1.1.21.2)$$

Then

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 4 & 2 \end{pmatrix} \quad (1.1.21.3)$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \quad (1.1.21.4)$$

Following row reduction,

$$\mathbf{A} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (1.1.21.5)$$

yielding (1.1.21.1).

1.1.22

$$|\mathbf{A}^\top| = |\mathbf{A}| \quad (1.1.22.1)$$

1.1.23

$$|\mathbf{AB}| = |\mathbf{A}| |\mathbf{B}| \quad (1.1.23.1)$$

Solution: If $\mathbf{A}$ is invertible

$$\mathbf{A} = \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k \quad (1.1.23.2)$$

where $\mathbf{E}_i$ are invertible matrices. For any single elementary matrix $\mathbf{E}$, it is easy to

verify that

$$|\mathbf{EB}| = |\mathbf{E}| |\mathbf{B}|. \quad (1.1.23.3)$$

$$\therefore |\mathbf{AB}| = |\mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k \mathbf{B}| \quad (1.1.23.4)$$

$$= |\mathbf{E}_1| |\mathbf{E}_2 \cdots \mathbf{E}_k \mathbf{B}| \quad (1.1.23.5)$$

$$= |\mathbf{E}_1| |\mathbf{E}_2| \cdots |\mathbf{E}_k| |\mathbf{B}| \quad (1.1.23.6)$$

For $\mathbf{B} = \mathbf{I}$,

$$|\mathbf{A}| = |\mathbf{E}_1| |\mathbf{E}_2| \cdots |\mathbf{E}_k| \quad (1.1.23.7)$$

which upon substituting in (1.1.23.6) yields

$$|\mathbf{AB}| = |\mathbf{A}| |\mathbf{B}| \quad (1.1.23.8)$$

1.1.24

$$|\mathbf{A}^{-1}| = \frac{1}{|\mathbf{A}|} \quad (1.1.24.1)$$

1.1.25 If

$$\mathbf{Ax} = \lambda \mathbf{x}, \quad (1.1.25.1)$$

then λ is an eigenvalue $\mathbf{x}$ of $\mathbf{A}$.

1.1.26 Let $\lambda_1, \lambda_2, \dots, \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then,

$$\sum_{i=1}^n \lambda_i = \text{Tr}(\mathbf{A}) \quad (1.1.26.1)$$

$$\prod_{i=1}^n \lambda_i = |\mathbf{A}| \quad (1.1.26.2)$$

1.1.27 $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.

Solution: From (1.1.24.1),

$$|\mathbf{A} - \lambda \mathbf{I}| = |(\mathbf{A} - \lambda \mathbf{I})^\top| \quad (1.1.27.1)$$

$$= |\mathbf{A}^\top - \lambda \mathbf{I}|, \quad (1.1.27.2)$$

completing the proof.

1.1.28 The equation of a conic with directrix $\mathbf{n}^\top \mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$g(\mathbf{x}) = \mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1.1.28.1)$$

where

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top, \quad (1.1.28.2)$$

$$\mathbf{u} = ce^2 \mathbf{n} - \|\mathbf{n}\|^2 \mathbf{F}, \quad (1.1.28.3)$$

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2 e^2 \quad (1.1.28.4)$$

Solution:

$$\|\mathbf{x} - \mathbf{F}\|^2 = e^2 \frac{(\mathbf{n}^\top \mathbf{x} - c)^2}{\|\mathbf{n}\|^2} \quad (1.1.28.5)$$

$$\implies \|\mathbf{n}\|^2 (\mathbf{x} - \mathbf{F})^\top (\mathbf{x} - \mathbf{F}) = e^2 (\mathbf{n}^\top \mathbf{x} - c)^2 \quad (1.1.28.6)$$

$$\implies \|\mathbf{n}\|^2 (\mathbf{x}^\top \mathbf{x} - 2\mathbf{F}^\top \mathbf{x} + \|\mathbf{F}\|^2) = e^2 (c^2 + (\mathbf{n}^\top \mathbf{x})^2 - 2c\mathbf{n}^\top \mathbf{x}) \quad (1.1.28.7)$$

$$= e^2 (c^2 + (\mathbf{x}^\top \mathbf{n} \mathbf{n}^\top \mathbf{x}) - 2c\mathbf{n}^\top \mathbf{x}) \quad (1.1.28.8)$$

which can be expressed as (1.1.28.1) after simplification.

1.1.29 The polar form of a conic section is given as:

$$r = \frac{ep}{1 + e \cos(\theta - \theta_0)} \quad (1.1.29.1)$$

where

- r is the distance from the focus(origin) to a point on the curve.
- e is the eccentricity of the conic
- p is the distance from the focus to the directrix
- θ_0 is the angle that defines the orientation of the major axis of the conic section relative to the polar reference axis ($\theta = 0$).

Solution: From (1.1.28.1), the general equation of a conic in matrix form is given by

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1.1.29.2)$$

For a conic with a focus at the origin ($\mathbf{F} = \mathbf{0}$), unit normal vector $\mathbf{n} = \begin{pmatrix} \cos \theta_0 \\ \sin \theta_0 \end{pmatrix}$, eccentricity e , and directrix distance parameter $c = -p$, substituting into (1.1.28.2)–(1.1.28.4) yields:

$$\mathbf{V} = \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top, \quad (1.1.29.3)$$

$$\mathbf{u} = -pe^2 \mathbf{n}, \quad (1.1.29.4)$$

$$f = -p^2 e^2 \quad (1.1.29.5)$$

Expressing $\mathbf{x}$ directly in polar coordinates yields

$$\mathbf{x} = r \mathbf{r}(\theta), \quad \text{where } \mathbf{r}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \quad (1.1.29.6)$$

Substituting into (1.1.29.2) yields

$$r^2 (\mathbf{r}(\theta)^\top \mathbf{V} \mathbf{r}(\theta)) + 2r (\mathbf{u}^\top \mathbf{r}(\theta)) + f = 0 \quad (1.1.29.7)$$

Evaluating the inner products:

$$\mathbf{n}^\top \mathbf{r}(\theta) = (\cos \theta_0 \quad \sin \theta_0) \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} = \cos(\theta - \theta_0) \quad (1.1.29.8)$$

$$\mathbf{r}(\theta)^\top \mathbf{V} \mathbf{r}(\theta) = \mathbf{r}(\theta)^\top (\mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top) \mathbf{r}(\theta) = 1 - e^2 \cos^2(\theta - \theta_0) \quad (1.1.29.9)$$

$$\mathbf{u}^\top \mathbf{r}(\theta) = -pe^2 \cos(\theta - \theta_0) \quad (1.1.29.10)$$

Substituting these into (1.1.29.7):

$$r^2 [1 - e^2 \cos^2(\theta - \theta_0)] - 2rpe^2 \cos(\theta - \theta_0) - p^2 e^2 = 0 \quad (1.1.29.11)$$

Factoring the quadratic expression:

$$[r(1 + e \cos(\theta - \theta_0)) - ep] [r(1 - e \cos(\theta - \theta_0)) + ep] = 0 \quad (1.1.29.12)$$

Taking the positive real root for radial distance $r > 0$:

$$r(1 + e \cos(\theta - \theta_0)) = ep \quad (1.1.29.13)$$

yielding (1.1.29.1)

1.1.30 Gravitational parameter of a body μ is defined as

$$\mu = GM \quad (1.1.30.1)$$

where M is the mass of the body

1.1.31 Specific angular momentum for a satellite with position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$ is

$$h = \|\mathbf{r} \times \mathbf{v}\| \quad (1.1.31.1)$$

1.1.32 Specific Energy for a satellite with position vector $\mathbf{r}$, velocity vector $\mathbf{v}$ and gravitational parameter μ is

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} \quad (1.1.32.1)$$

where

$$r = \|\mathbf{r}\|, v = \|\mathbf{v}\| \quad (1.1.32.2)$$

1.1.33 The gravitational acceleration $\mathbf{a}$ acting on a satellite is given by Newton's law of gravitation:

$$\mathbf{a} = -\frac{\mu}{r^3} \mathbf{r} \quad (1.1.33.1)$$

1.1.34 The trajectory of a satellite under a central gravitational force is always a conic section.

Solution: Taking the cross product of $\mathbf{r}$ with $\mathbf{a}$:

$$\mathbf{r} \times \mathbf{a} = \mathbf{r} \times \left(-\frac{\mu}{r^3} \mathbf{r} \right) = \mathbf{0} \quad (1.1.34.1)$$

From (1.1.34.1)

$$\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \mathbf{v} \times \mathbf{v} + \mathbf{r} \times \mathbf{a} = \mathbf{0} \quad (1.1.34.2)$$

Hence, the specific angular momentum vector $\mathbf{h}$ is conserved:

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = \text{constant} \quad (1.1.34.3)$$

Crossing the acceleration $\mathbf{a}$ with $\mathbf{h}$:

$$\mathbf{a} \times \mathbf{h} = \left(-\frac{\mu}{r^3}\mathbf{r}\right) \times (\mathbf{r} \times \mathbf{v}) \quad (1.1.34.4)$$

$$= -\frac{\mu}{r^3} [(\mathbf{r} \cdot \mathbf{v})\mathbf{r} - (\mathbf{r} \cdot \mathbf{r})\mathbf{v}] \quad (1.1.34.5)$$

$$= \frac{\mu}{r^3} [r^2\mathbf{v} - (\mathbf{r} \cdot \mathbf{v})\mathbf{r}] \quad (1.1.34.6)$$

Using $\mathbf{r} \cdot \mathbf{v} = r \frac{dr}{dt}$, this simplifies to:

$$\mathbf{a} \times \mathbf{h} = \mu \left[\frac{\mathbf{v}}{r} - \frac{1}{r^2} \frac{dr}{dt} \mathbf{r} \right] = \mu \frac{d}{dt} \left(\frac{\mathbf{r}}{r} \right) \quad (1.1.34.7)$$

Integrating with respect to time t :

$$\mathbf{v} \times \mathbf{h} = \mu \left(\frac{\mathbf{r}}{r} + \mathbf{C} \right) \quad (1.1.34.8)$$

where $\mathbf{C}$ is the integration constant. Taking the scalar product of (1.1.34.8) with $\mathbf{r}$:

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = \mu \mathbf{r} \cdot \left(\frac{\mathbf{r}}{r} + \mathbf{C} \right) \quad (1.1.34.9)$$

$$\implies (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mu (r + rC \cos \theta) \quad (1.1.34.10)$$

$$\implies \|\mathbf{h}\|^2 = \mu r (1 + C \cos \theta) \quad (1.1.34.11)$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{C}$. Rearranging for r yields

$$r = \frac{\frac{\|\mathbf{h}\|^2}{\mu}}{1 + C \cos \theta} = \frac{ep}{1 + e \cos \theta} \quad (1.1.34.12)$$

This is the same equation as (1.1.29.1) where

$$ep = \frac{\|\mathbf{h}\|^2}{\mu} \quad (1.1.34.13)$$

$$\mathbf{e} = \mathbf{C} \quad (1.1.34.14)$$

1.1.35 The squared eccentricity e^2 of a satellite's trajectory is related to its specific orbital energy ε and specific angular momentum magnitude h by

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \quad (1.1.35.1)$$

Solution: From (1.1.34.8) and (1.1.34.14),

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \quad (1.1.35.2)$$

Taking the magnitude squared of both sides using $\|\mathbf{e}\|^2 = \mathbf{e} \cdot \mathbf{e}$:

$$e^2 = \|\mathbf{e}\|^2 = \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \right) \cdot \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \right) \quad (1.1.35.3)$$

Expanding the inner product:

$$e^2 = \frac{\|\mathbf{v} \times \mathbf{h}\|^2}{\mu^2} - \frac{2}{\mu r} [\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h})] + \frac{\mathbf{r} \cdot \mathbf{r}}{r^2} \quad (1.1.35.4)$$

We know that

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mathbf{h} \cdot \mathbf{h} = h^2 \quad (1.1.35.5)$$

And,

$$\mathbf{r} \cdot \mathbf{r} = r^2 \quad (1.1.35.6)$$

Substituting these back into (1.1.35.4) yields

$$e^2 = \frac{v^2 h^2}{\mu^2} - \frac{2h^2}{\mu r} + 1 \quad (1.1.35.7)$$

$$= 1 + \frac{2h^2}{\mu^2} \left(\frac{v^2}{2} - \frac{\mu}{r} \right) \quad (1.1.35.8)$$

Substituting (1.1.32.1) into (1.1.35.8) yields:

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \quad (1.1.35.9)$$

1.1.36 Consider the system A system comprising a bar, spring and mass is shown in the figure Fig. 1.1.36.1 The bar, having negligible mass, is made of a material having Young's modulus E , cross-sectional area A and length L . The spring stiffness is k and the mass is M kg. Find the axial stiffness of the bar.

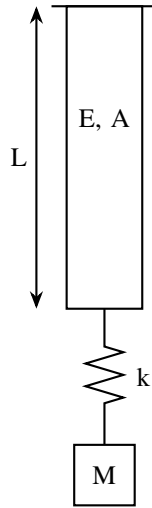


Fig. 1.1.36.1. Spring mass system

Solution: The axial stiffness of the bar is

$$k_b = \frac{AE}{L} \quad (1.1.36.1)$$

1.1.37 Find the overall spring constant in Fig. 1.1.36.1.

Solution: Using Hooke's law, by balancing the displacements,

$$\frac{Mg}{k_t} = \frac{Mg}{k_b} + \frac{Mg}{k} \quad (1.1.37.1)$$

$$\Rightarrow k_{eq} = \frac{k_b k}{k_b + k} \quad (1.1.37.2)$$

where k_t is the overall spring constant.

1.1.38 Show that the equation for motion for the mass in Fig. 1.1.38.1 is

$$\frac{d^2 x}{dt^2} = -\frac{k}{M}x \quad (1.1.38.1)$$

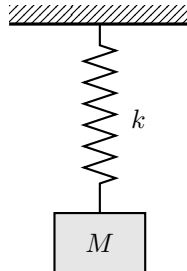


Fig. 1.1.38.1. Spring Mass system in vertical plane

Solution: When the mass M is at rest, let the system stretch by δ_{st} . Then,

$$Mg - k\delta_{st} = 0 \implies Mg = k\delta_{st} \quad (1.1.38.2)$$

When the mass is displaced downward by $x(t)$, using Hooke's law,

$$M \frac{d^2x}{dt^2} = Mg - k(\delta_{st} + x) \quad (1.1.38.3)$$

$$(1.1.38.4)$$

yielding (1.1.38.1), upon substituting from (1.1.38.2).

2. BT 2025

A. Formulae

2.1.1 Exponential Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad (2.1.1.1)$$

2.1.2 A circle with center $\mathbf{c}$ and radius r , can be expressed as (1.1.28.1) with

$$\mathbf{V} = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = -\mathbf{c}, \quad f = \|\mathbf{c}\|^2 - r^2 \quad (2.1.2.1)$$

Solution: The equation of a circle is

$$\|\mathbf{x} - \mathbf{c}\|^2 = r^2 \quad (2.1.2.2)$$

$$(\mathbf{x} - \mathbf{c})^\top (\mathbf{x} - \mathbf{c}) = r^2 \quad (2.1.2.3)$$

$$\mathbf{x}^\top \mathbf{x} - 2\mathbf{c}^\top \mathbf{x} + \mathbf{c}^\top \mathbf{c} - r^2 = 0 \quad (2.1.2.4)$$

yielding (2.1.2.1) upon comparing with (1.1.28.1)

2.1.3 (Line-Conic Intersection) The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by

$$\mathbf{X} = (\mathbf{h} \quad \mathbf{m}) \begin{pmatrix} 1 & 1 \\ \kappa_1 & \kappa_2 \end{pmatrix} \quad (2.1.3.1)$$

where

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}}, i = 1, 2. \quad (2.1.3.2)$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V} \mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$.

2.1.4 (Intersection of Conics and Common Chord) The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ is given by the pencil:

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \quad (2.1.4.1)$$

The value of μ corresponding to the degenerate conic (the radical axis / common chord line) is found from

$$\left| \begin{array}{cc} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{array} \right| = 0 \quad (2.1.4.2)$$

2.1.5 Laplace Transform

- **Bilateral Definition:** For an arbitrary continuous-time signal $g(t)$, the bilateral Laplace transform is defined as:

$$G(s) = \mathcal{L}\{g(t)\} = \int_{-\infty}^{\infty} g(t) e^{-st} dt \quad (2.1.5.1)$$

- **Unilateral Definition:** For causal signals and differential equations with initial conditions at $t = 0$:

$$G(s) = \mathcal{L}\{g(t)\} = \int_0^{\infty} g(t) e^{-st} dt \quad (2.1.5.2)$$

- **Derivation of Derivative Properties:** Applying integration by parts to $\mathcal{L}\{y'(t)\}$ with $u = e^{-st}$ and $dv = y'(t) dt \implies du = -se^{-st} dt$, $v = y(t)$:

$$\mathcal{L}\{y'(t)\} = \int_0^{\infty} y'(t) e^{-st} dt \quad (2.1.5.3)$$

$$= [y(t) e^{-st}]_0^{\infty} - \int_0^{\infty} y(t) (-se^{-st}) dt \quad (2.1.5.4)$$

$$= (0 - y(0)) + s \int_0^{\infty} y(t) e^{-st} dt \quad (2.1.5.5)$$

$$= sY(s) - y(0) \quad (2.1.5.6)$$

Thus, the first derivative transform is:

$$y'(t) \xleftrightarrow{\mathcal{L}} sY(s) - y(0) \quad (2.1.5.7)$$

Applying the first derivative property recursively to the second derivative $y''(t) = \frac{d}{dt}(y'(t))$:

$$\mathcal{L}\{y''(t)\} = s\mathcal{L}\{y'(t)\} - y'(0) \quad (2.1.5.8)$$

$$= s[sY(s) - y(0)] - y'(0) \quad (2.1.5.9)$$

$$= s^2Y(s) - sy(0) - y'(0) \quad (2.1.5.10)$$

Under zero initial conditions ($y(0) = 0$, $y'(0) = 0$), the operational transforms simplify to:

$$y(t) \xleftrightarrow{\mathcal{L}} Y(s) \quad (2.1.5.11)$$

$$y'(t) \xleftrightarrow{\mathcal{L}} sY(s) \quad (2.1.5.12)$$

$$y''(t) \xleftrightarrow{\mathcal{L}} s^2Y(s) \quad (2.1.5.13)$$

- **Transform of the Unit Step Function:** The Laplace transform of an unconstrained

constant $f(t) = 1$ over $(-\infty, \infty)$ is undefined because the integral diverges as $t \rightarrow -\infty$. For causal systems, a constant input starting at $t = 0$ is defined through the unit step function $u(t)$:

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} \quad (2.1.5.14)$$

Evaluating the unilateral Laplace transform of $u(t)$:

$$\mathcal{L}\{u(t)\} = \int_0^{\infty} 1 \cdot e^{-st} dt = \left[-\frac{e^{-st}}{s} \right]_0^{\infty} = \frac{1}{s}, \quad \text{Re}(s) > 0 \quad (2.1.5.15)$$

Hence:

$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s} \quad (2.1.5.16)$$

- Transform of e^{at} :

$$e^{at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s-a}, \quad \text{Re}(s) > a \quad (2.1.5.17)$$

$$(2.1.5.18)$$

2.1.6 Euler's Method of Forward Differences

- For a first-order differential equation $\frac{dy}{dt} = f(t, y)$ with step size h :

$$y_{n+1} = y_n + hf(t_n, y_n) \quad (2.1.6.1)$$

2.1.7 Gradient Descent Optimization

- **First-Order Parameter Update:** For a differentiable function $f(x)$, the iterative parameter update with learning rate $\eta > 0$ is given by:

$$x_{k+1} = x_k - \eta f'(x_k) \quad (2.1.7.1)$$

- **Second Derivative Test for Convexity:**

$$f''(x) > 0 \quad \forall x \in (0, \infty) \implies f(x) \text{ is strictly convex} \quad (2.1.7.2)$$

A strictly convex function on an open interval possesses at most one local minimum, which is also its unique global minimum.

2.1.8 Minimization via Completing the Square

- For any $x > 0$ and constant $a > 0$, expanding the squared difference gives:

$$\left(\sqrt{x} - \frac{a}{\sqrt{x}} \right)^2 = x - 2a + \frac{a^2}{x} \quad (2.1.8.1)$$

Because the square of any real term is non-negative $\left(\sqrt{x} - \frac{a}{\sqrt{x}} \right)^2 \geq 0$, rearranging yields the lower bound:

$$x + \frac{a^2}{x} = \left(\sqrt{x} - \frac{a}{\sqrt{x}} \right)^2 + 2a \geq 2a \quad (2.1.8.2)$$

The equality holds if and only if the squared term vanishes:

$$\sqrt{x} - \frac{a}{\sqrt{x}} = 0 \implies x = a \quad (2.1.8.3)$$

attaining the unique minimum value $2a$.

2.1.9 Newton-Raphson Method

- First-Order Recurrence (Difference Equation): For a real-valued differentiable function $f(x)$, the root is iteratively approximated using its first-order Taylor series expansion about the current estimate x_n :

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (2.1.9.1)$$

2.1.10 System of Linear Equations and Rouché–Capelli Theorem

- A system of m linear equations in n variables can be represented in matrix form as:

$$\mathbf{Ax} = \mathbf{b} \quad (2.1.10.1)$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the coefficient matrix, $\mathbf{x} \in \mathbb{R}^{n \times 1}$ is the vector of unknowns, and $\mathbf{b} \in \mathbb{R}^{m \times 1}$ is the constant vector.

- The solvability of the system is governed by the ranks of the coefficient matrix $\mathbf{A}$ and the augmented matrix $[\mathbf{A} \mid \mathbf{b}]$:
 - a) Inconsistent System (No Solution): A linear system has no solution if and only if:

$$\text{rank}(\mathbf{A}) < \text{rank}([\mathbf{A} \mid \mathbf{b}]) \quad (2.1.10.2)$$

Geometrically, this corresponds to distinct parallel lines that do not intersect.

- b) Consistent System (At Least One Solution): A linear system has at least one solution if and only if:

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \quad (2.1.10.3)$$

- If $r = n$ (where n is the number of variables), there exists a unique solution.
- If $r < n$, there exist infinitely many solutions with $n - r$ free variables (coincident lines).

2.1.11 Sum of Independent Discrete Random Variables (Two Dice Roll)

- Let $X_1, X_2 \in \{1, 2, 3, 4, 5, 6\}$ be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF):

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{6}, & 1 \leq n \leq 6 \\ 0, & \text{otherwise} \end{cases} \quad (2.1.11.1)$$

- Let $X = X_1 + X_2$ denote their sum. The PMF of X is given by the discrete

convolution of $p_{X_1}(n)$ and $p_{X_2}(n)$:

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k \Pr(X_1 = n - k \mid X_2 = k) p_{X_2}(k) \quad (2.1.11.2)$$

$$= \sum_k p_{X_1}(n - k) p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \quad (2.1.11.3)$$

- Substituting (2.1.11.1) into (2.1.11.3) yields the triangular PMF:

$$p_X(n) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \leq n \leq 7 \\ \frac{13-n}{36}, & 7 < n \leq 12 \\ 0, & n > 12 \end{cases} \quad (2.1.11.4)$$

2.1.12 Unit Step Response of an RC Circuit and s -Domain Analysis

- Time-Domain Circuit Equation: Consider a series RC circuit excited by a step input $v_i(t) = V_{dc}u(t)$ where V_{dc} is a constant DC voltage source, and $u(t)$ is step input function defined as:

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} \quad (2.1.12.1)$$

Starting at $t = 0$, the current through the capacitor is:

$$i(t) = \frac{dq}{dt} = C \frac{dV_{out}}{dt} \quad (2.1.12.2)$$

Applying Kirchhoff's Voltage Law (KVL) around the loop:

$$Ri(t) + V_{out}(t) = V_{dc} \implies RC \frac{dV_{out}}{dt} + V_{out}(t) = V_{dc} \quad (2.1.12.3)$$

- s -Domain Representation: Transforming the circuit to the Laplace domain, the step source becomes $V_{in}(s) = \frac{V_{dc}}{s}$, the resistor impedance is R , and the capacitor impedance is $\frac{1}{sC}$. The equivalent impedance is:

$$R_{eq}(s) = R + \frac{1}{sC} = \frac{RsC + 1}{sC} \quad (2.1.12.4)$$

The loop current in the s -domain is:

$$I(s) = \frac{V_{in}(s)}{R_{eq}(s)} = \frac{\frac{V_{dc}}{s}}{\frac{RsC + 1}{sC}} = \frac{V_{dc}C}{RsC + 1} \quad (2.1.12.5)$$

The output voltage across the capacitor is obtained as:

$$V_{out}(s) = I(s) \left(\frac{1}{sC} \right) = \frac{V_{dc}}{s(RCs + 1)} \quad (2.1.12.6)$$

$$= \frac{V_{dc}}{RC} \left[\frac{1}{s \left(s + \frac{1}{RC} \right)} \right] = V_{dc} \left[\frac{1}{s} - \frac{1}{s + \frac{1}{RC}} \right] \quad (2.1.12.7)$$

Taking the inverse Laplace transform yields:

$$V_{out}(t) = V_{dc} \left(1 - e^{-\frac{t}{RC}}\right) u(t) \quad (2.1.12.8)$$

2.1.13 Standard First-Order Differential Equation Comparison

- A general first-order dynamic system with time constant τ subjected to a unit step input is described by:

$$\tau \frac{dy}{dt} + y(t) = y_{ss} \quad (2.1.13.1)$$

where τ is the time constant and y_{ss} is the steady-state response.

- Comparing (2.1.13.1) with the RC circuit in (2.1.12.3):

$$y(t) \longleftrightarrow V_{out}(t), \quad \tau \longleftrightarrow RC, \quad y_{ss} \longleftrightarrow V_{dc} \quad (2.1.13.2)$$

Transforming (2.1.13.1) into the s -domain with zero initial conditions $y(0) = 0$:

$$\tau s Y(s) + Y(s) = \frac{y_{ss}}{s} \implies Y(s) = \frac{y_{ss}}{s(\tau s + 1)} = y_{ss} \left[\frac{1}{s} - \frac{1}{s + \frac{1}{\tau}} \right] \quad (2.1.13.3)$$

Taking the inverse Laplace transform directly gives:

$$y(t) = y_{ss} \left(1 - e^{-\frac{t}{\tau}}\right) u(t) \quad (2.1.13.4)$$

2.1.14 Transfer Function of a First-Order Process

- In continuous-time signals and linear time-invariant (LTI) systems, the transfer function $H(s)$ is defined as the ratio of the Laplace transform of the output signal $Y(s)$ to the Laplace transform of the input signal $X(s)$, under the condition that all initial conditions are zero:

$$H(s) = \frac{Y(s)}{X(s)} \quad (2.1.14.1)$$

- For a process driven by an input forcing function $g(t)$ with Laplace transform $G(s)$, the transfer function is written as:

$$H(s) = \frac{Y(s)}{G(s)} \quad (2.1.14.2)$$

where $G(s)$ denotes the input function and $Y(s)$ is the process output response evaluated under relaxed (zero) initial conditions.

- In the time domain, this corresponds to the convolution of the input signal with the system impulse response $h(t) = \mathcal{L}^{-1}\{H(s)\}$:

$$y(t) = h(t) * g(t) \quad (2.1.14.3)$$

2.1.15 Identity Condition for Linear Combinations

- For any independent variables or functions x and y , if the relation

$$ax + by = 0 \quad (2.1.15.1)$$

holds identically for all x and y , then the coefficients must vanish simultaneously:

$$a = 0 \quad \text{and} \quad b = 0 \quad (2.1.15.2)$$

- Similarly, if an expression $C_1 f_1(t) + C_2 f_2(t) = 0$ holds for arbitrary constants C_1 and C_2 , each coefficient must vanish independently:

$$f_1(t) = 0 \quad \text{and} \quad f_2(t) = 0 \quad (2.1.15.3)$$

For distinct exponential functions $e^{r_1 t}$ and $e^{r_2 t}$ ($r_1 \neq r_2$), this implies linear independence across all t .

2.1.16 Continuity and Differentiability in Functions

- **Continuity at a Point:** A real-valued function $f(x)$ is continuous at $x = x_0$ if and only if the left-hand limit, the right-hand limit, and the value of the function coincide:

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = f(x_0) \quad (2.1.16.1)$$

- **Differentiability and Slope Matching:** A function $f(x)$ is differentiable at $x = x_0$ if the left-hand derivative equals the right-hand derivative:

$$f'(x_0^-) = \lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} = f'(x_0^+) \quad (2.1.16.2)$$

For piecewise functions defined by differentiable curves on each side of the boundary x_0 , this requires matching boundary slopes:

$$f'(x_0^-) = f'(x_0^+) \quad (2.1.16.3)$$

- **Differentiability Implies Continuity:** If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$:

$$\lim_{x \rightarrow x_0} [f(x) - f(x_0)] = \lim_{x \rightarrow x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} (x - x_0) \right] = f'(x_0) \cdot 0 = 0 \quad (2.1.16.4)$$

Hence, continuity at $x = x_0$ is a necessary prerequisite for differentiability.

2.1.17 Piecewise Boundary Conditions at $x = a$

- Consider a function $f(x)$ defined by differentiable branches $f_1(x)$ and $f_2(x)$ across a boundary point $x = a$:

$$f(x) = \begin{cases} f_1(x), & x \leq a \\ f_2(x), & x > a \end{cases} \quad (2.1.17.1)$$

For $f(x)$ to be continuous at $x = a$, the one-sided limits must equal the boundary value:

$$\lim_{x \rightarrow a^-} f_1(x) = \lim_{x \rightarrow a^+} f_2(x) \implies f_1(a) = f_2(a) \quad (2.1.17.2)$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives at the boundary must coincide:

$$f'_1(a) = f'_2(a) \quad (2.1.17.3)$$

2.1.18 Representation of Linear Balance Systems as Matrix Equations

- A general system of m linear equations in n variables $x_1, x_2, \dots, x_n$:

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i, \quad i = 1, 2, \dots, m \quad (2.1.18.1)$$

can be expressed in matrix-vector form:

$$\mathbf{A}\mathbf{x} = \mathbf{b} \quad (2.1.18.2)$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the coefficient matrix, $\mathbf{x} \in \mathbb{R}^n$ is the state vector, and $\mathbf{b} \in \mathbb{R}^m$ is the constant vector:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad (2.1.18.3)$$

- In multi-component conservation and stoichiometric balances, the conservation of elemental species across reactant and product streams is represented by:

$$\mathbf{A}_{\text{reactants}}\mathbf{x}_{\text{reactants}} = \mathbf{A}_{\text{products}}\mathbf{x}_{\text{products}} \quad (2.1.18.4)$$

Combining both sides into a single stoichiometric matrix yields the homogeneous system:

$$[\mathbf{A}_{\text{reactants}} \quad -\mathbf{A}_{\text{products}}]\mathbf{x} = \mathbf{0} \quad (2.1.18.5)$$

When boundary parameters or specific yields fix certain degrees of freedom (such as basis mole values and yield coefficients), the homogeneous system reduces to a deterministic non-homogeneous matrix equation $\mathbf{A}\mathbf{x} = \mathbf{b}$ that can be solved directly via row operations or symbolic matrix inversion.

2.1.19 Solution of First-Order Recurrence Relations via Z-Transform

- The bilateral Z-transform of a discrete-time sequence $x[n]$ is defined as:

$$X(z) = \mathcal{Z}[x[n]] = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad (2.1.19.1)$$

over the Region of Convergence (ROC) where the series converges absolutely.

- Discrete Unit Step Sequence: The discrete unit step function $u[n]$ is defined as:

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (2.1.19.2)$$

Its forward Z-transform is given by:

$$\mathcal{Z}[u[n]] = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}}, \quad |z| > 1 \quad (2.1.19.3)$$

- **Time-Shift Property:** Shifting a sequence by an integer delay k corresponds to multiplication by z^{-k} in the Z-domain:

$$\mathcal{Z}[x[n - k]] = z^{-k} X(z) \quad (2.1.19.4)$$

Consequently, delaying the discrete unit step by one sample yields:

$$\mathcal{Z}[u[n - 1]] = \frac{z^{-1}}{1 - z^{-1}}, \quad |z| > 1 \quad (2.1.19.5)$$

- **Standard Inverse Transform Pair:** For any scalar constant α :

$$\mathcal{Z}^{-1}\left[\frac{1}{1 - \alpha z^{-1}}\right] = \alpha^n u[n], \quad |z| > |\alpha| \quad (2.1.19.6)$$

2.1.20 Relation Between Eigenvalues and Trace of a Matrix

- Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. The characteristic polynomial of $\mathbf{A}$ is defined by:

$$P(\lambda) = |\lambda \mathbf{I} - \mathbf{A}| \quad (2.1.20.1)$$

- Factoring the polynomial in terms of its n characteristic roots:

$$P(\lambda) = \prod_{i=1}^n (\lambda - \lambda_i) = \lambda^n - \left(\sum_{i=1}^n \lambda_i\right) \lambda^{n-1} + \dots + (-1)^n \prod_{i=1}^n \lambda_i \quad (2.1.20.2)$$

- Evaluating the determinant $|\lambda \mathbf{I} - \mathbf{A}|$ using cofactor expansion, the only term that contributes to λ^n and λ^{n-1} is the product of the principal diagonal entries:

$$\prod_{i=1}^n (\lambda - a_{ii}) = \lambda^n - \left(\sum_{i=1}^n a_{ii}\right) \lambda^{n-1} + \dots \quad (2.1.20.3)$$

where a_{ii} denotes the diagonal elements of $\mathbf{A}$.

- Equating the coefficient of λ^{n-1} from (2.1.20.2) and (2.1.20.3):

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} = \text{tr}(\mathbf{A}) \quad (2.1.20.4)$$

Hence, the sum of all eigenvalues of a square matrix is always equal to its trace.

B. Problems

2.2.1 The sum of the following infinite series is:

$$\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \cdots \quad (2.2.1.1)$$

(BT 2025)

a) π

b) $1 + e$

c) $e - 1$

d) e

Solution: (2.2.1.1) can be expressed as

$$\sum_{n=1}^{\infty} \frac{1}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} - 1 = e - 1 \quad (2.2.1.2)$$

upon substituting from (2.1.1.1) The following C code verifies the series sum numerically.

codes/btq3/btq3.c

2.2.2 A circle with center at $(x, y) = (0.5, 0)$ and radius $r_1 = 0.5$ intersects with another circle with center at $(x, y) = (1, 1)$ and radius $r_2 = 1$ at two points. One of the points of intersection (x, y) is: (BT 2025)

a) $(0, 0)$

b) $(0.2, 0.4)$

c) $(0.5, 0.5)$

d) $(1, 2)$

Solution: From (2.1.2.1), the circle parameters are

$$\mathbf{c}_1 = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix}, r_1 = 0.5, \quad \mathbf{u}_1 = -\mathbf{c}_1 = \begin{pmatrix} -0.5 \\ 0 \end{pmatrix}, \quad f_1 = \|\mathbf{c}_1\|^2 - r_1^2 = 0 \quad (2.2.2.1)$$

$$\mathbf{c}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, r_2 = 1, \quad \mathbf{u}_2 = -\mathbf{c}_2 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \quad f_2 = \|\mathbf{c}_2\|^2 - r_2^2 = 1 \quad (2.2.2.2)$$

Using the pencil of conics from (2.1.4.1), the condition for a degenerate conic from (2.1.4.2) gives:

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \quad (2.2.2.3)$$

Substituting the parameter matrices:

$$\begin{vmatrix} 1 + \mu & 0 & -0.5 - \mu \\ 0 & 1 + \mu & -\mu \\ -0.5 - \mu & -\mu & \mu \end{vmatrix} = 0 \quad (2.2.2.4)$$

Expanding the determinant:

$$(1 + \mu) [(1 + \mu) \mu - \mu^2] - (-0.5 - \mu) [0 - (1 + \mu) (-0.5 - \mu)] = 0 \quad (2.2.2.5)$$

$$(1 + \mu) \mu - (1 + \mu) (\mu + 0.5)^2 = 0 \quad (2.2.2.6)$$

$$(1 + \mu) \left(-\mu^2 - \frac{1}{4} \right) = 0 \quad (2.2.2.7)$$

Since $\mu^2 + \frac{1}{4} = 0$ yields imaginary roots, the only real root is:

$$\mu = -1 \quad (2.2.2.8)$$

Substituting $\mu = -1$ into (2.1.4.1) eliminates the second-degree terms ($\mathbf{V}_1 - \mathbf{V}_2 = \mathbf{0}$), yielding the common chord (radical axis):

$$2(\mathbf{u}_1 - \mathbf{u}_2)^\top \mathbf{x} + (f_1 - f_2) = 0 \quad (2.2.2.9)$$

$$2 \left(\begin{pmatrix} -0.5 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ -1 \end{pmatrix} \right)^\top \mathbf{x} + (0 - 1) = 0 \quad (2.2.2.10)$$

$$2 \begin{pmatrix} 0.5 & 1 \end{pmatrix} \mathbf{x} - 1 = 0 \implies \begin{pmatrix} 1 & 2 \end{pmatrix} \mathbf{x} = 1 \quad (2.2.2.11)$$

Thus, the equation of the line passing through the intersection points is $x + 2y = 1$. The normal vector of this line is $\mathbf{n} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. A direction vector $\mathbf{m}$ orthogonal to $\mathbf{n}$ ($\mathbf{m}^\top \mathbf{n} = 0$) and a point $\mathbf{h}$ on the line are:

$$\mathbf{m} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{h} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (2.2.2.12)$$

Expressing the line in parametric form $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ and substituting into Circle 1 using (2.1.3.2):

$$\mathbf{m}^\top \mathbf{V}_1 \mathbf{m} = \begin{pmatrix} 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = 5 \quad (2.2.2.13)$$

$$\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -0.5 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} \quad (2.2.2.14)$$

$$\mathbf{m}^\top (\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1) = \begin{pmatrix} 2 & -1 \end{pmatrix} \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} = 1 \quad (2.2.2.15)$$

$$g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}_1 \mathbf{h} + 2\mathbf{u}_1^\top \mathbf{h} + f_1 = 1 - 1 + 0 = 0 \quad (2.2.2.16)$$

Evaluating the parameter roots κ :

$$\kappa = \frac{-1 \pm \sqrt{1^2 - 0}}{5} = \frac{-1 \pm 1}{5} \implies \kappa_1 = 0, \quad \kappa_2 = -\frac{2}{5} \quad (2.2.2.17)$$

The points of intersection are:

$$\mathbf{x}_1 = \mathbf{h} + 0\mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (2.2.2.18)$$

$$\mathbf{x}_2 = \mathbf{h} - \frac{2}{5}\mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{2}{5} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0.2 \\ 0.4 \end{pmatrix} \quad (2.2.2.19)$$

Hence, one of the points of intersection is (0.2, 0.4), and option (B) is correct.

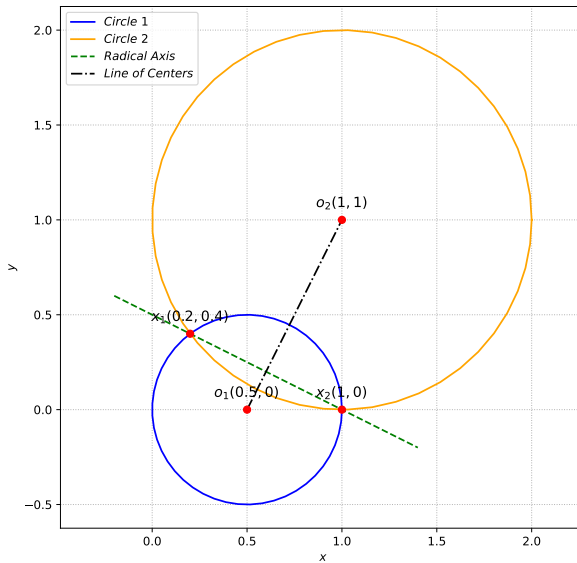


Fig. 2.2.2.1. Intersection of the two circles and the radical axis

codes/btq9/circleProblem1.py

figs/circleintersect.pdf

2.2.3 Let $y(t)$ be a bacterial population whose growth is given by:

$$\frac{dy}{dt} = \lambda(y + 2) \quad (2.2.3.1)$$

where λ is the growth rate constant. If $y(0) = 1$ and $y(1) = 4$, then the value of λ is: (BT 2025)

- a) $\ln 2$
- b) $\ln 3$
- c) $\ln 4$
- d) $\ln 6$

Solution: Rearranging the differential equation into standard linear form:

$$\frac{dy}{dt} - \lambda y = 2\lambda \quad (2.2.3.2)$$

Taking the unilateral Laplace transform on both sides using (2.1.5.7) and (2.1.5.15):

$$sY(s) - y(0) - \lambda Y(s) = \frac{2\lambda}{s} \quad (2.2.3.3)$$

Substituting the initial condition $y(0) = 1$:

$$(s - \lambda) Y(s) - 1 = \frac{2\lambda}{s} \quad (2.2.3.4)$$

$$(s - \lambda) Y(s) = 1 + \frac{2\lambda}{s} = \frac{s + 2\lambda}{s} \quad (2.2.3.5)$$

$$Y(s) = \frac{s + 2\lambda}{s(s - \lambda)} \quad (2.2.3.6)$$

Decomposing into partial fractions:

$$\frac{s + 2\lambda}{s(s - \lambda)} = \frac{A}{s} + \frac{B}{s - \lambda} \implies s + 2\lambda = A(s - \lambda) + Bs \quad (2.2.3.7)$$

2.2.4 Evaluating the coefficients:

$$\text{At } s = 0 : \quad 2\lambda = A(-\lambda) \implies A = -2 \quad (2.2.4.1)$$

$$\text{At } s = \lambda : \quad 3\lambda = B(\lambda) \implies B = 3 \quad (2.2.4.2)$$

Substituting A and B back into $Y(s)$ and applying the inverse Laplace transform using (2.1.5.15) and (2.1.5.17):

$$Y(s) = -\frac{2}{s} + \frac{3}{s - \lambda} \quad (2.2.4.3)$$

$$y(t) = 3e^{\lambda t} - 2 \quad (2.2.4.4)$$

Applying the boundary condition $y(1) = 4$:

$$4 = 3e^{\lambda(1)} - 2 \quad (2.2.4.5)$$

$$6 = 3e^{\lambda} \quad (2.2.4.6)$$

$$e^{\lambda} = 2 \implies \lambda = \ln 2 \quad (2.2.4.7)$$

Hence, option (A) is correct.

Numerical Simulation (Euler's Method): Applying the recurrence relation from (2.1.6.1) with step size $h = 0.01$ and $\lambda = \ln 2$:

$$y_{n+1} = y_n + 0.01 [\ln(2)(y_n + 2)], \quad y_0 = 1 \quad (2.2.4.8)$$

The C code implementation is as follows:

codes/btq13/bac.c

For all $x \in (0, \infty)$, $f''(x) > 0$. Thus, $f(x)$ is strictly convex, and $x = 2$ corresponds to a unique global minimum.

Evaluating the function at $x = 2$:

$$f(2) = 2 + \frac{4}{2} = 4 \quad (2.2.5.6)$$

Hence, option (D) is correct.

Numerical Simulation (Gradient Descent): Applying the first-order update relation from (2.1.7.1) with learning rate $\eta = 0.3$:

$$x_{k+1} = x_k - 0.3 \left(1 - \frac{4}{x_k^2} \right) \quad (2.2.5.7)$$

Starting from the initial guess $x_0 = 5$, the sequence converges to $x^* = 2$, yielding $f(x^*) = 4$.

2.2.6 Alternative Method (Completing the Square): For $x > 0$, the function $f(x)$ can be expressed in terms of square roots and rewritten as a perfect square:

$$f(x) = x + \frac{4}{x} \quad (2.2.6.1)$$

$$= (\sqrt{x})^2 - 2(\sqrt{x}) \left(\frac{2}{\sqrt{x}} \right) + \left(\frac{2}{\sqrt{x}} \right)^2 + 2(\sqrt{x}) \left(\frac{2}{\sqrt{x}} \right) \quad (2.2.6.2)$$

$$= \left(\sqrt{x} - \frac{2}{\sqrt{x}} \right)^2 + 4 \quad (2.2.6.3)$$

Since the square of any real number is non-negative:

$$\left(\sqrt{x} - \frac{2}{\sqrt{x}} \right)^2 \geq 0 \quad (2.2.6.4)$$

with equality holding if and only if:

$$\sqrt{x} - \frac{2}{\sqrt{x}} = 0 \implies \sqrt{x} = \frac{2}{\sqrt{x}} \implies x = 2 \quad (2.2.6.5)$$

Therefore:

$$f(x) \geq 4 \quad (2.2.6.6)$$

The minimum value is 4, attained at $x = 2$, which confirms option (D).

The C code implementing iterative gradient descent is:

```
codes/btq14/min.c
```

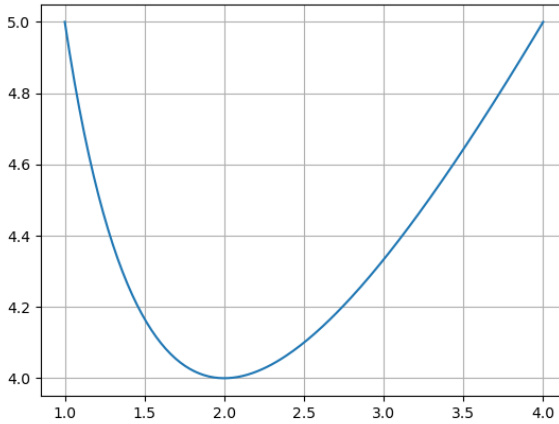


Fig. 2.2.6.1. Plot of $f(x) = x + \frac{4}{x}$ indicating the minimum at $(2, 4)$

codes/btq14/min.py

figs/min.png

2.2.7 Consider a nonlinear algebraic equation, $e^x - 2 = 0$. Using the Newton-Raphson method, with the initial guess of $x_0 = 1$, the approximated value of the root of the equation after one iteration is _____ (round off to two decimal places). (BT 2025)

Solution: The given objective function and its first derivative are:

$$f(x) = e^x - 2 \quad (2.2.7.1)$$

$$f'(x) = e^x \quad (2.2.7.2)$$

Substituting $f(x)$ and $f'(x)$ into the Newton-Raphson recurrence relation from (2.1.9.1):

$$x_{n+1} = x_n - \frac{e^{x_n} - 2}{e^{x_n}} \quad (2.2.7.3)$$

$$= x_n - \left(1 - \frac{2}{e^{x_n}}\right) \quad (2.2.7.4)$$

$$= x_n - 1 + 2e^{-x_n} \quad (2.2.7.5)$$

Given the initial guess $x_0 = 1$, the first iteration ($n = 0$) gives:

$$x_1 = x_0 - 1 + 2e^{-x_0} \quad (2.2.7.6)$$

$$= 1 - 1 + 2e^{-1} \quad (2.2.7.7)$$

$$= \frac{2}{e} \approx \frac{2}{2.71828} \approx 0.73576 \approx 0.74 \quad (2.2.7.8)$$

Subsequent Iterations (Convergence to $\ln 2$):

$$x_2 = x_1 - 1 + 2e^{-x_1} \approx 0.694042 \quad (2.2.7.9)$$

$$x_3 = x_2 - 1 + 2e^{-x_2} \approx 0.693148 \quad (2.2.7.10)$$

$$x_4 = x_3 - 1 + 2e^{-x_3} \approx 0.693147 \approx \ln 2 \quad (2.2.7.11)$$

The exact mathematical solution is $e^x = 2 \implies x = \ln 2 \approx 0.693147$. Thus, the approximated value of the root after one iteration is 0.74.

The C Code implementing Newton-Raphson recurrence is as follows:

codes/btq33/raph.c

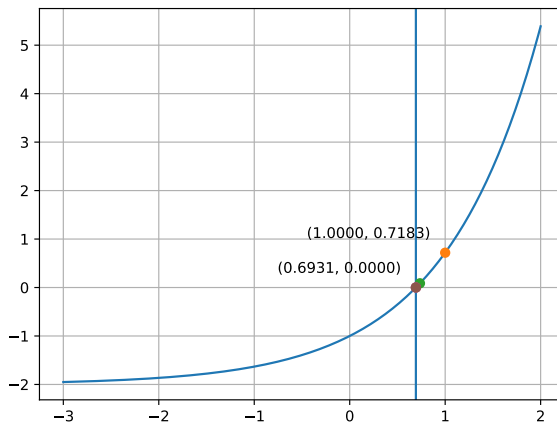


Fig. 2.2.7.1. Convergence of the Newton-Raphson method to $x = \ln 2$

codes/btq33/raph.py

2.2.8 The value of k , for which the linear equations $2x + 3y = 6$ and $4x + 6y = 3k$ have at least one solution, is _____ (answer in integer). (BT 2025)

Solution: The given system of linear equations is:

$$2x + 3y = 6 \quad (2.2.8.1)$$

$$4x + 6y = 3k \quad (2.2.8.2)$$

Expressing in matrix form $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 3k \end{pmatrix} \quad (2.2.8.3)$$

The augmented matrix is formulated as:

$$(\mathbf{A} \quad \mathbf{b} \mid) = \left(\begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 3k \end{array} \right) \quad (2.2.8.4)$$

Applying Gaussian elimination using the elementary row operation $R_2 \rightarrow R_2 - 2R_1$:

$$\left(\begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 3k \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{cc|c} 2 & 3 & 6 \\ 0 & 0 & 3k - 12 \end{array} \right) \quad (2.2.8.5)$$

The coefficient matrix $\mathbf{A}$ reduces to:

$$\mathbf{A} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{cc} 2 & 3 \\ 0 & 0 \end{array} \right) \quad (2.2.8.6)$$

Since there is only one non-zero row in echelon form, $\text{rank}(\mathbf{A}) = 1$.

From the Rouché–Capelli theorem in (2.1.10.3), for the system to have at least one solution, it must be consistent:

$$\text{rank}((\mathbf{A} \quad \mathbf{b} \mid)) = \text{rank}(\mathbf{A}) = 1 \quad (2.2.8.7)$$

This requires the second row of the augmented matrix to be entirely zeros:

$$3k - 12 = 0 \quad (2.2.8.8)$$

$$3k = 12 \implies k = 4 \quad (2.2.8.9)$$

For $k = 4$, $\text{rank}(\mathbf{A}) = \text{rank}((\mathbf{A} \quad \mathbf{b} \mid)) = 1 < 2$ (number of variables $n = 2$), yielding infinitely many solutions along the coincident line $2x + 3y = 6$.

Hence, the integer value of k is 4.

The Python Code showing the implementation for a given value of k and comparing solutions is as follows:

codes/btq34/bt34.py

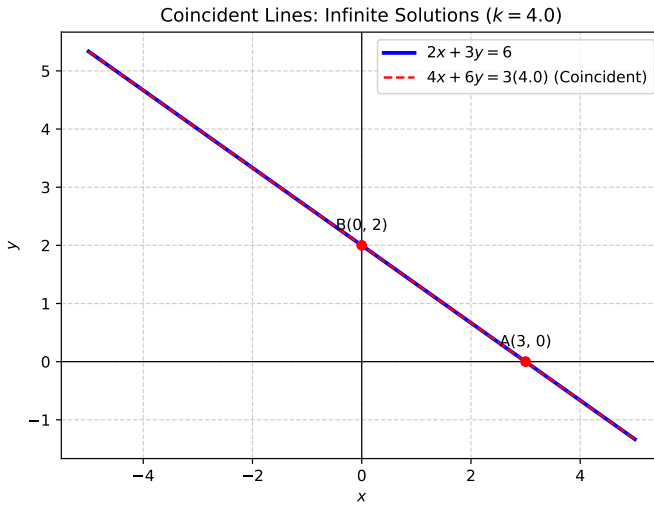


Fig. 2.2.8.1. Coincident lines representing infinitely many solutions at $k = 4$

2.2.9 Two fair six-sided dice, with sides numbered 1 to 6, are thrown once. The probability of getting 7 as the sum of the numbers on the top side of the dice is _____. (BT 2025)

Solution: Let X_1 and X_2 represent the scores on the two dice, and let $X = X_1 + X_2$ denote their sum. From the triangular distribution derived in (2.1.11.4), the probability mass function for $2 \leq n \leq 7$ is given by:

$$p_X(n) = \frac{n-1}{36} \quad (2.2.9.1)$$

Evaluating the PMF at $n = 7$:

$$\Pr(X = 7) = \frac{7-1}{36} = \frac{6}{36} = \frac{1}{6} \quad (2.2.9.2)$$

Hence, option (A) is correct.

The Python simulation script verifying the PMF across 10,000 trials is:

codes/btq35/dice.py

The instrument reaches 95% of its steady-state reading when $y(t_{95}) = 0.95y_{ss}$:

$$y_{ss} \left(1 - e^{-\frac{t_{95}}{\tau}}\right) = 0.95y_{ss} \quad (2.2.10.4)$$

$$1 - e^{-\frac{t_{95}}{\tau}} = 0.95 \quad (2.2.10.5)$$

$$e^{-\frac{t_{95}}{\tau}} = 0.05 = \frac{1}{20} \quad (2.2.10.6)$$

Taking the natural logarithm on both sides:

$$-\frac{t_{95}}{\tau} = \ln(0.05) = -\ln(20) \quad (2.2.10.7)$$

$$t_{95} = \tau \ln(20) \quad (2.2.10.8)$$

Substituting the given time constant $\tau = 40$ s:

$$t_{95} = 40 \ln(20) \quad (2.2.10.9)$$

$$\approx 40 \times 2.99573 \approx 119.83 \text{ s} \quad (2.2.10.10)$$

Rounding off to the nearest integer gives 120 seconds. Hence, option (C) is correct.

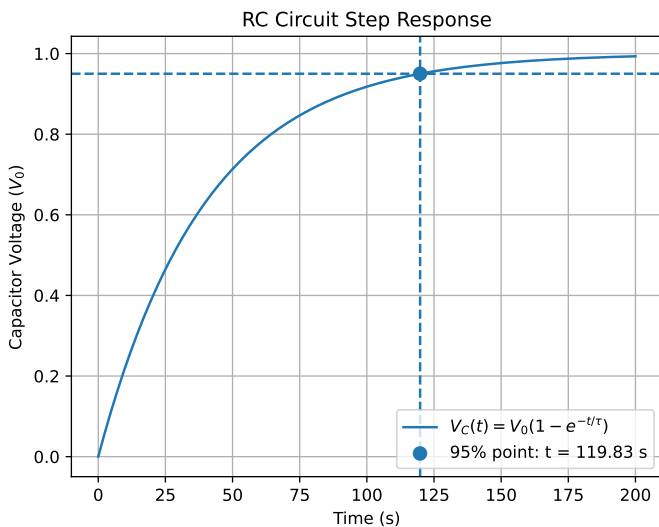


Fig. 2.2.10.1. Step response of the first-order system with $\tau = 40$ s

codes/btq42/plot.py

figs/rc-step-response.pdf

2.2.11 The output $y(t)$ of a first-order process is governed by the following differential equation

$$\tau_p \frac{dy}{dt} + y = K_p f(t) \quad (2.2.11.1)$$

where τ_p is a non-zero time constant, K_p is the gain and $f(t)$ is the input with $f(0) = 0$.

Assume $y(0) = 0$. The transfer function for this process is (consider s as the independent variable in the Laplace domain) (BT 2025)

- | | |
|-------------------------------|------------------------------|
| a) $\frac{K_p}{\tau_p s + 1}$ | c) $\frac{\tau_p}{K_p(s+1)}$ |
| b) $\frac{\tau_p}{K_p s + 1}$ | d) $\frac{K_p}{\tau_p(s+1)}$ |

Solution: The given first-order differential equation governing the process is:

$$\tau_p \frac{dy(t)}{dt} + y(t) = K_p f(t) \quad (2.2.11.2)$$

Taking the unilateral Laplace transform on both sides using (2.1.5.7):

$$\tau_p [sY(s) - y(0)] + Y(s) = K_p F(s) \quad (2.2.11.3)$$

Applying the given initial condition $y(0) = 0$:

$$\tau_p sY(s) + Y(s) = K_p F(s) \quad (2.2.11.4)$$

$$Y(s)(\tau_p s + 1) = K_p F(s) \quad (2.2.11.5)$$

The transfer function is the ratio of the output $Y(s)$ to the input $F(s)$:

$$\frac{Y(s)}{F(s)} = \frac{K_p}{\tau_p s + 1} \quad (2.2.11.6)$$

Hence, option (A) is correct.

2.2.12 Let m and n be fixed real numbers. If the function $y(t) = C_1 e^t + C_2 e^{-t}$ is a solution of

$$\frac{d^2 y}{dt^2} + m \frac{dy}{dt} + ny = 0 \quad (2.2.12.1)$$

for any constants C_1 and C_2 , then $m + n$ is equal to (BT 2025)

- | | |
|-------|------|
| a) -2 | c) 0 |
| b) -1 | d) 1 |

Solution: The given function is:

$$y(t) = C_1 e^t + C_2 e^{-t} \quad (2.2.12.2)$$

Computing the first and second derivatives with respect to t :

$$\frac{dy}{dt} = C_1 e^t - C_2 e^{-t} \quad (2.2.12.3)$$

$$\frac{d^2 y}{dt^2} = C_1 e^t + C_2 e^{-t} \quad (2.2.12.4)$$

Substituting y , $\frac{dy}{dt}$, and $\frac{d^2 y}{dt^2}$ into the differential equation:

$$\frac{d^2 y}{dt^2} + m \frac{dy}{dt} + ny = 0 \quad (2.2.12.5)$$

$$(C_1 e^t + C_2 e^{-t}) + m (C_1 e^t - C_2 e^{-t}) + n (C_1 e^t + C_2 e^{-t}) = 0 \quad (2.2.12.6)$$

Grouping the terms containing C_1 and C_2 :

$$C_1 e^t (1 + m + n) + C_2 e^{-t} (1 - m + n) = 0 \quad (2.2.12.7)$$

Using (2.1.15.1) and (2.1.15.2), since this equation must hold for any constants C_1 and C_2 :

$$1 + m + n = 0 \quad (2.2.12.8)$$

$$1 - m + n = 0 \quad (2.2.12.9)$$

Adding (2.2.12.8) and (2.2.12.9):

$$2 + 2n = 0 \implies n = -1 \quad (2.2.12.10)$$

Subtracting (2.2.12.9) from (2.2.12.8):

$$2m = 0 \implies m = 0 \quad (2.2.12.11)$$

Evaluating $m + n$:

$$m + n = 0 + (-1) = -1 \quad (2.2.12.12)$$

Hence, option (B) is correct.

2.2.13 Consider the piecewise function

$$f(x) = \begin{cases} \sin(2x), & x > 0 \\ a + bx, & x \leq 0 \end{cases} \quad (2.2.13.1)$$

where a and b are real constants. If $f(x)$ is differentiable at $x = 0$, then the value of $a + b$ is (BT 2025)

- | | |
|------|------|
| a) 0 | c) 2 |
| b) 1 | d) 3 |

Solution: Since $f(x)$ is differentiable at $x = 0$, from (2.1.16.4) it must also be continuous at $x = 0$. Applying the continuity condition from (2.1.16.1):

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = f(0) \quad (2.2.13.2)$$

Evaluating the one-sided limits:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \sin(2x) = \sin(0) = 0 \quad (2.2.13.3)$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (a + bx) = a + b(0) = a \quad (2.2.13.4)$$

Equating the limits yields:

$$a = 0 \quad (2.2.13.5)$$

Applying the differentiability condition and matching one-sided derivatives at $x = 0$ using (2.1.16.3):

$$f'(0^+) = f'(0^-) \quad (2.2.13.6)$$

For $x > 0$:

$$f'(x) = \frac{d}{dx} (\sin(2x)) = 2 \cos(2x) \implies f'(0^+) = 2 \cos(0) = 2 \quad (2.2.13.7)$$

For $x < 0$:

$$f'(x) = \frac{d}{dx} (a + bx) = b \implies f'(0^-) = b \quad (2.2.13.8)$$

Equating the two slopes:

$$b = 2 \quad (2.2.13.9)$$

Evaluating $a + b$:

$$a + b = 0 + 2 = 2 \quad (2.2.13.10)$$

Hence, option (C) is correct.

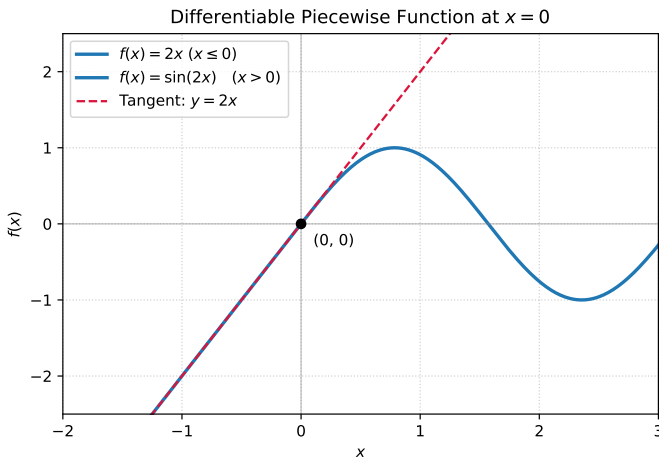
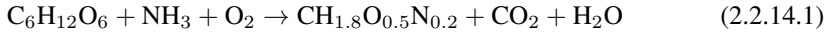


Fig. 2.2.13.1. Differentiable piecewise function and its tangent line $y = 2x$ at $x = 0$

codes/btq46/bt46.py

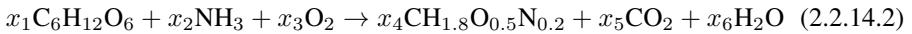
figs/piecewise-tangent-plot.pdf

2.2.14 In a fermentation process, each mole of glucose is converted to biomass ($\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$), with a biomass yield coefficient of 0.4 C-mol/C-mol, according to the unbalanced equation given below.



The moles of oxygen consumption per mole of glucose consumed during fermentation is _____. (BT 2025)

Solution: Let the balanced stoichiometric reaction be represented as:



where $x_1, x_2, x_3, x_4, x_5, x_6$ are the respective stoichiometric coefficients.

Formulating the elemental conservation balances for C, N, H, and O using exact rational fractions:

$$\text{Carbon (C): } 6x_1 = x_4 + x_5 \quad (2.2.14.3)$$

$$\text{Nitrogen (N): } x_2 = \frac{1}{5}x_4 \quad (2.2.14.4)$$

$$\text{Hydrogen (H): } 12x_1 + 3x_2 = \frac{9}{5}x_4 + 2x_6 \quad (2.2.14.5)$$

$$\text{Oxygen (O): } 6x_1 + 2x_3 = \frac{1}{2}x_4 + 2x_5 + x_6 \quad (2.2.14.6)$$

Writing the elemental balances in matrix form $\mathbf{A}_{\text{LHS}}\mathbf{x}_{\text{reactants}} = \mathbf{A}_{\text{RHS}}\mathbf{x}_{\text{products}}$:

$$\begin{pmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 12 & 3 & 0 \\ 6 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ \frac{1}{5} & 0 & 0 \\ 0 & 2 \\ \frac{1}{2} & 2 & 1 \end{pmatrix} \begin{pmatrix} x_4 \\ x_5 \\ x_6 \end{pmatrix} \quad (2.2.14.7)$$

Constructing the stoichiometric matrix $\mathbf{A} = [\mathbf{A}_{\text{LHS}} \quad -\mathbf{A}_{\text{RHS}}]$:

$$\begin{pmatrix} 6 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & -\frac{1}{5} & 0 & 0 \\ 12 & 3 & 0 & -\frac{9}{5} & 0 & -2 \\ 6 & 0 & 2 & -\frac{1}{2} & -2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (2.2.14.8)$$

The biomass yield coefficient is 0.4 C-mol/C-mol = $\frac{2}{5}$ C-mol/C-mol, which provides the relation:

$$x_4 - \frac{2}{5}(6x_1) = 0 \implies x_4 = \frac{12}{5}x_1 \quad (2.2.14.9)$$

Solving the system symbolically in terms of x_4 :

$$x_1 = \frac{5}{12}x_4 \quad (2.2.14.10)$$

$$x_2 = \frac{1}{5}x_4 \quad (2.2.14.11)$$

$$x_3 = \frac{29}{20}x_4 \quad (2.2.14.12)$$

$$x_5 = \frac{3}{2}x_4 \quad (2.2.14.13)$$

$$x_6 = \frac{19}{10}x_4 \quad (2.2.14.14)$$

Evaluating with the sample numerical basis $x_4 = 5.0$:

$$x_1 = \frac{25}{12} \approx 2.0833 \quad (2.2.14.15)$$

$$x_2 = 1.0000 \quad (2.2.14.16)$$

$$x_3 = 7.2500 \quad (2.2.14.17)$$

$$x_4 = 5.0000 \quad (2.2.14.18)$$

$$x_5 = 7.5000 \quad (2.2.14.19)$$

$$x_6 = 9.5000 \quad (2.2.14.20)$$

The moles of oxygen consumed per mole of glucose consumed during fermentation is given by:

$$\frac{x_3}{x_1} = \frac{\frac{29}{20}x_4}{\frac{5}{12}x_4} = \frac{29}{20} \times \frac{12}{5} = \frac{87}{25} = 3.48 \quad (2.2.14.21)$$

or using the numerical evaluation:

$$\frac{x_3}{x_1} = \frac{7.2500}{2.0833} = 3.48 \quad (2.2.14.22)$$

Rounding off to two decimal places, the answer is 3.48.

The implementation in SymPy to find solutions is as follows:

codes/btq58/58.py

2.2.15 Let $a_0 = 0$ and define $a_n = \frac{1}{2}(1 + a_{n-1})$ for all positive integers $n \geq 1$. The least value of n for which $|1 - a_n| < \frac{1}{2^{10}}$ is _____.

(BT 2025)

Solution: The given recurrence relation is:

$$a_n = \frac{1}{2}(1 + a_{n-1}), \quad n \geq 1, \quad a_0 = 0 \quad (2.2.15.1)$$

Extending the difference equation across all $n \in \mathbb{Z}$ by causal multiplication with $u[n]$ from (2.1.19.2):

$$a_n u[n] - \frac{1}{2} a_{n-1} u[n-1] = \frac{1}{2} u[n-1] \quad (2.2.15.2)$$

Let $A(z) = \mathcal{Z}[a_n u[n]]$. Applying the time-shift property (2.1.19.4) and the delayed step transform (2.1.19.5):

$$A(z) - \frac{1}{2}z^{-1}A(z) = \frac{1}{2}\mathcal{Z}[u[n-1]] \quad (2.2.15.3)$$

$$A(z) \left(1 - \frac{1}{2}z^{-1}\right) = \frac{\frac{1}{2}z^{-1}}{1 - z^{-1}} \quad (2.2.15.4)$$

Solving algebraically for $A(z)$:

$$A(z) = \frac{\frac{1}{2}z^{-1}}{(1 - z^{-1})(1 - \frac{1}{2}z^{-1})}, \quad |z| > 1 \quad (2.2.15.5)$$

Performing partial fraction decomposition with $w = z^{-1}$:

$$\frac{\frac{1}{2}w}{(1-w)(1-\frac{1}{2}w)} = \frac{C_1}{1-w} + \frac{C_2}{1-\frac{1}{2}w} \quad (2.2.15.6)$$

where

$$C_1 = \frac{\frac{1}{2}w}{1 - \frac{1}{2}w} \Big|_{w=1} = \frac{1/2}{1/2} = 1 \quad (2.2.15.7)$$

$$C_2 = \frac{\frac{1}{2}w}{1 - w} \Big|_{w=2} = \frac{1}{1-2} = -1 \quad (2.2.15.8)$$

Substituting back into $A(z)$:

$$A(z) = \frac{1}{1 - z^{-1}} - \frac{1}{1 - \frac{1}{2}z^{-1}} \quad (2.2.15.9)$$

Applying the inverse Z-transform using (2.1.19.3) and (2.1.19.6):

$$a_n = \left[1 - \left(\frac{1}{2}\right)^n\right] u[n] = 1 - 2^{-n}, \quad n \geq 0 \quad (2.2.15.10)$$

Applying the stopping threshold condition $|1 - a_n| < \frac{1}{2^{10}}$:

$$|1 - (1 - 2^{-n})| < \frac{1}{2^{10}} \quad (2.2.15.11)$$

$$2^{-n} < 2^{-10} \quad (2.2.15.12)$$

$$2^n > 2^{10} \implies n > 10 \quad (2.2.15.13)$$

Since n must be an integer, the minimal integer satisfying the strict inequality is:

$$n = 11 \quad (2.2.15.14)$$

The implementation in C to simulate the recurrence and verify the threshold is as follows:

codes/btq59/ztransform.c

2.2.16 Let $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & k & 0 \\ 3 & 0 & -1 \end{pmatrix}$. If the eigenvalues of A are -2 , 1 , and 2 , then the value of k is _____.

(BT 2025)

Solution: The given matrix is:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & k & 0 \\ 3 & 0 & -1 \end{pmatrix} \quad (2.2.16.1)$$

The eigenvalues of $\mathbf{A}$ are given as:

$$\lambda_1 = -2, \quad \lambda_2 = 1, \quad \lambda_3 = 2 \quad (2.2.16.2)$$

From (2.1.20.4), the sum of the eigenvalues of a square matrix is equal to its trace:

$$\sum_{i=1}^3 \lambda_i = \text{tr}(\mathbf{A}) \quad (2.2.16.3)$$

Computing the trace of $\mathbf{A}$:

$$\text{tr}(\mathbf{A}) = 1 + k + (-1) = k \quad (2.2.16.4)$$

Computing the sum of the eigenvalues:

$$\sum_{i=1}^3 \lambda_i = (-2) + 1 + 2 = 1 \quad (2.2.16.5)$$

Equating the two expressions:

$$k = 1 \quad (2.2.16.6)$$

The solution can be verified from the following Python script:

codes/btq63/eigen.py